Nikita Jayaprakash

Berkeley, CA | (925) 353-6120 | nikita.jp@gmail.com | https://github.com/nikita-jaya | www.linkedin.com/in/nikita-jaya

EDUCATION

Ph.D. in Demography | University of California, Berkeley 2026 - PRESENT
Interests: infectious diseases, social networks, statistical techniques

Masters of Science in Applied Data Science | Carnegie Mellon University 2025 - 2026

Bachelor's of Arts in Statistics | University of California, Berkeley 2022 - 2025
Cluster in Mathematics, Minors in Demography and English

Pembroke Cambridge Summer Programme | University of Cambridge Summer 2023

SKILLS & QUALIFICATIONS

- **Programming Languages:**
  - Python (pandas, numpy, matplotlib, xgboost, scikit-learn, pytorch, streamlit)
  - R (dplyr, tidyr, lubridate, ggplot2, caret, xgboost, randomForest, purrr, Shiny)
- **Additional Tools**: SQL, GitHub, LaTex, Microsoft Suite (Office, Excel, Powerpoint)
- **Technical Skills:** practical applications of statistics and AI, data-driven best practices, machine learning, analytical problem-solving
- **Soft Skills:** interpersonal skills, written and oral communication skills, public speaking, organized, leadership skills, creativity
- **Domain Experience:** academic research, education, academic writing (IMRaD), open-source software, nonprofit environments

ACADEMIC EXPERIENCE

Graduate Student Researcher **Aug 2026 - PRESENT**
Department of Demography, UC Berkeley Berkeley, CA
- Conduct research related to the dynamics of infectious diseases and social networks under the direction of a faculty member

Senior AI Curriculum Fellow **July 2026 - PRESENT**
D-Lab, UC Berkeley Berkeley, CA
- Facilitate AI literacy workshops and supported D-Lab programming for the broader Berkeley research community
- Develop new workshop curriculum related to AI education and workforce development for a diverse audience

Research Assistant **June 2026 - Aug 2026**
NSF AI Institute for Societal Decision Making, Carnegie Mellon University Pittsburgh, PA
- Developed products for emergency response in a partnership between CMU's AI-SDM and Meta's AI for Good program, utilizing Meta's limited-access mobility datasets and computer vision models (SAM) for disaster analysis to inform first responders
- Produced real-time dynamic situation reports for natural disasters like earthquakes, hurricanes and severe winter storms

AI Fellow **Aug 2025 - May 2026**
NSF AI Institute for Societal Decision Making, Carnegie Mellon University Pittsburgh, PA
- Received inaugural NSF AI-SDM Fellowships for CMU master's of applied data science students
- Presented at and attended workshops for 50+ hybrid participants (for industry and general public) on generative Al Literacy

Research Assistant **Aug 2025 - May 2026**
Department of Statistics, UC Berkeley Berkeley, CA
- Assisted in experiment that compares grading accuracy across experts (faculty), student graders and multimodal LLMs (AI)
- Developed workflow to streamline study design while maintaining student privacy (CITI program: Social & Behavioral Research)

Data Science Student Researcher **Sept 2024 - Dec 2024**
Aspiranet Berkeley, CA
- Presented in collaboration with 5 students about therapy attendance of foster youth at UC Berkeley's Discovery Symposium
- Utilized data analytics and statistical software to identify patterns in cancellation behavior over 3 years of data of 2000+ clients
- Advised representative on effectiveness of text message reminders in reducing therapy appointment cancellations among youth

Student Researcher **Feb 2023 - May 2025**
Department of Statistics, UC Berkeley Berkeley, CA
- Co-founded *NemoGB*, an instructor-facing learning management system deployed on main Berkeley software repository
- Received $34,000 grant from Statistics Department, impacting ~10,000 students across large data science and statistics courses
- Presented live demos to professors in order to demonstrate grading functionality and receive user feedback during development

PUBLICATIONS

- **Jayaprakash, Nikita**, Voutov, Monika, Turner, Zach, and Andrew Bray. NemoGB: An R Package for Calculating Course Grades. *Journal of Open Source Education*, 8(94), 303. https://doi.org/10.21105/jose.00303.

MANUSCRIPTS UNDER REVIEW

- Bray, Andrew, Grossman, Josh, **Jayaprakash, Nikita**, and Mine Çetinkaya-Rundel. Grading the Graders: Comparing AI and Human Grading Accuracy for Short Answer Questions. *Journal of Statistics and Data Science Education*.

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PROFESSIONAL EXPERIENCE

Development Research Intern **Jan 2025 - June 2025**
Major Giving Department, KQED San Francisco, CA
- Conducted qualitative research of 100+ donor profiles and strategic event briefings for Development and Executive Management
- Developed an interactive 15-tab Excel spreadsheet of 1200+ donors for cross-portfolio comparison to guide business strategy

Data, Audits & Contracts Intern **June 2024 - Aug 2024**
Education & Employment Services Department, Fortune Society Queens, NY
- Audited data about reported demographics, job placements and client assessments in order to assess contract compliance
- Analyzed 1450+ formerly incarcerated clients' data (e.g. demographics, career milestones) over the past two years that resulted in a focused effort to address areas of outreach to increase program retention

AI / Data Science / ML / Developer Intern **Jan 2024 - May 2024**
Hyphenova Remote
- Led exploratory data analysis team in guiding decision-making with quantitative analysis of 700+ social media creators
- Built a Shiny app (in Python) to allow for quick analysis of creators in order to assess accuracy of statistical modeling

TEACHING EXPERIENCE

High School Instructor **June 2026 - July 2026**
Academic Talent Development Program, UC Berkeley Berkeley, CA
- Taught two 6-week summer courses (Applied Statistics in R; Data Science with R) to advanced high schoolers
- Developed the entire curriculum for Applied Statistics in R to introduce statistics through the lens of experimental design

High School Instructor **June 2025 - July 2025**
Academic Talent Development Program, UC Berkeley Berkeley, CA
- Taught 6-week summer course to 16 advanced high schoolers via UC Berkeley's Academic Talent Development Program
- Modified curriculum to include a final project using R Shiny while maintaining STEM course and program objectives
- Employed flipped-classroom learning environment to maximize hands-on coding experience with real-world datasets

Stat 20 Group Tutor **Jan 2023 - May 2025**
Department of Statistics, UC Berkeley Berkeley, CA
- Appointed to help students via Statistics Scholar program (intended to assist students of diverse backgrounds within the course) by communicating with students, planning group tutoring sessions and conducting quiz reviews
- Supported 100+ students and coordinated with professor in interactive lectures and assignments in an introductory statistics course

ACADEMIC PROJECTS

Vooks Behavioral Signals in User Engagement Capstone Project **Spring 2026**
- Predicted trial conversion and subscription churn with machine learning models (LASSO logistic regression, XGBoost, balanced random forest) and survival analysis techniques (time-varying cox proportional hazard model) for a storybook streaming platform
- Presented actionable business insights based on stakeholder input, data-driven recommendations, and qualitative industry research

Deep Learning and Clustering Analysis of Gene Expression Project **Spring 2026**
- Utilized applications of deep learning (variational autoencoders), statistical genomics, and high-dimensional inference
- Partition the 3005 cells into biologically meaningful groups that align with known cell types and generated low-dimensional embeddings that provide insight into the underlying structure and separability of the single-cell RNA sequencing of mice

U.S. Weekly Gasoline Prices Time Series Project **Spring 2026**
- Utilized vector autoregressive methods in order to analyze the relationship of American gasoline prices with crude oil prices and comparing regional gasoline prices within the United States in order to model, forecast and understand US gasoline price dynamics

ETL (Extract, Transform, Load) Data Pipeline Project **Fall 2025**
• Designed database structure and table schema for PostgreSQL database design with a team of 2 other graduate students using large data sets from Health and Human Services and hospital quality information from Centers for Medicare and Medicaid Services
- Develop reports and data analytics using Python's Streamlit to summarize data with benchmarking

Goodreads Textual Analysis Project **Fall 2025**
- Developed a web scraper to scrape 2713 Goodreads descriptions of books banned in the United States from 2023 to 2024
- Wrote a full IDMRaD report that used multi-dimensional factor analysis of the Biber features and keyness in order to determine literary features and literary differences between the Goodreads descriptions of more-banned and less-banned books

HIV Treatment Data Analysis Project **Fall 2025**
- Utilized linear regression models to analyze 2139 observations from a randomized clinical trial for HIV treatment data
- Investigated prolonged clinical benefit from drug intervention and responses to HIV therapy according to progression of disease and prior HIV drug exposure using observational causal inference methodologies